A Spatiotemporal Graph Neural Network for Nationwide Daily District-Level PM2.5 Estimation in Bangladesh

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FINAL YEAR THESIS REPORT

This Report Presented in Partial Fulfillment of the Requirements for the **Degree of Bachelor of Science in Computer Science and Engineering**

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APPROVAL

This Project titled "**A Spatiotemporal Graph Neural Network for Nationwide Daily District-Level PM2.5 Estimation in Bangladesh**," submitted by **Kh Sadman Sakib** and **Wahid Tousif Somoy** to the Department of Computer Science and Engineering, Daffodil International University, has been accepted as satisfactory for the partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on **12-07-2026**.

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We hereby declare that this project has been done by us under the supervision of **Md. Ferdouse Ahmed Foysal, Lecturer,** Department of Computer Science and Engineering, Daffodil International University. We also declare that neither this projectnor any part of this project has been submitted elsewhere for the award of any degree or diploma.

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ABSTRACT

Even though Bangladesh ranks among the most polluted countries in the world, due to the lack of existing ground stations, it has no openly available high-resolution PM2.5 record covering the whole country. Therefore, most works in the field are based around the country's capital Dhaka, with advanced ML models like Spatio-Temporal Graph Neural Network (ST-GNN) never being applied. We address both gaps. First, we construct the first daily, district-level PM2.5 dataset covering all 64 districts of Bangladesh over 2019-2025 (7 years, 163,584 district-day records), fusing satellite remote sensing (Sentinel-5P TROPOMI, MODIS MAIAC), meteorological reanalysis (ERA5-Land), and ground measurements from the OpenAQ and Department of Environment networks, with a Random Forest calibration (held-out $R^2 = 0.64$) filling the gaps where no measurement exists, yielding a complete dataset in which every value is flagged as measured or predicted. In terms of PM2.5, only 8.45% of values are direct measurements and 41 out of 64 districts are never monitored, so we evaluate all predictions against an independent satellite product (ACAG), which agrees closely even in unmeasured districts ($r = 0.84$) showing evidence that the gap-filled values are trustworthy. Second, we establish the first ever ST-GNN benchmark for the region by comparing baselines, sequence models, and four established ST-GNN architectures under a three-track evaluation protocol that separates well-observed, unmonitored (zero-shot), and independently-validated regimes. Building on these comparisons, we propose a spatiotemporal model that replaces fixed graph adjacencies with a learned self-attention mechanism over districts. This enables the model to discover spatial dependencies directly from data and in unmonitored districts it yields a statistically significant improvement (zero-shot R^2 0.761 vs. 0.753, $p < 0.05$, ten seeds), although, on well-observed districts it matches the sequence baseline ($R^2 \approx 0.83$). This confirms that learned spatial structure adds value precisely where ground truth is absent. However, in terms of zero-shot accuracy, the strongest models converge near 0.76, nothing pushes past it.

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Introduction

This chapter introduces the problem the project addresses, the motivation behind it, the specific objectives pursued, a high-level summary of the methodology, the outcomes achieved, and the organization of the remainder of the report.

1.1 Introduction

Ambient air pollution, in particular, particulate matter with a diameter of 2.5 micrometers or less (PM2.5), is among the most severe environmental health risks of the present time. PM2.5 is small enough to travel deep into the alveoli and the finest fraction can cross into the bloodstream. Chronic exposure to PM2.5. is causally associated with cardiovascular disease, respiratory illness, and premature mortality **Error! Reference source not found.**. The World Health Organization's 2021 air quality guidelines set a 24-hour mean PM2.5 limit of 15 micrograms per cubic meter, a threshold that much of South Asia exceeds on the large majority of days each year [2].

By most international assessment, Bangladesh is among the most polluted countries in the world, which regularly appears at or near the top of global PM2.5 rankings [3]. Despite the significance of the problem, the country's capacity to measure and model its own air quality is sharply limited. Ground-based monitoring is sparse and unevenly distributed. Bangladesh's Department of Environment tracks only a few dozen regulatory monitoring stations that operate across a country of 64 districts and roughly 170 million people, and most of these stations were commissioned only in the last few years. Because of this, the record of PM2.5 across Bangladesh is fragmentary, and this patchiness creates two separate problems.

The first is a **data problem**. Before this work, there was no openly available, high-resolution, multi-year record of daily PM2.5 at the district level for Bangladesh. The second is a **modeling problem** that comes straight out of the first. With so few measurements, no spatiotemporal machine-learning model — and in particular no spatiotemporal graph neural network (ST-GNN), the kind of model now widely used for air-quality prediction in other countries — had ever been built or tested for Bangladesh. And the issue is not just that these models had never been tried here. It is that the way they are normally built does not fit this setting. ST-GNNs are usually developed on dense monitoring networks where almost every location has its own sensor, and they usually fix the graph linking locations ahead of time, based on distance or wind direction. In Bangladesh, where most districts have never been measured at all, neither of those assumptions holds.

This project tackles both problems together. We construct a daily, district-level PM2.5 dataset covering all 64 districts of Bangladesh over the seven-year period from 2019 to 2025, fusing satellite remote sensing, meteorological reanalysis, and the limited available ground measurements into a spatially and temporally complete field with transparency. Our main modeling contribution is a spatiotemporal graph neural network built for exactly this kind of data-sparse setting. Instead of fixing the graph between districts from distance or wind, the proposed model **learns** that graph from the data, using a self-attention mechanism over districts that works out on its own which districts inform which. The reason for this design is simple, and the results back it up, in a country where most districts have no history of their own and have to borrow information from others, the districts that actually help each other are not always the nearest ones — so it is better to let the model find those links than to assume them.

Put simply, the problem this work addresses has two parts: how do you build a complete and trustworthy PM2.5 record for a country with almost no monitoring, and how do you then predict pollution across that country, most of which is unmonitored, using a model that learns spatial structure instead of being told it? The dataset answers the first part. The proposed model addresses the second part. The model is developed and tested within the first proper ST-GNN benchmark created for the region.

1.2 Motivation

The motivation comes from both human and technical factors. For humans, air pollution significantly affects public health in Bangladesh. You cannot create effective policy without understanding where the pollution actually occurs. The country has never had a complete, day-by-day view of PM2.5 levels outside Dhaka. As a result, exposure studies, health research, and policy decisions have relied on a few stations to represent a nation of 64 districts. Creating a daily, district-level record is important by itself. It provides researchers and policymakers with a consistent national record to use, offering a level of detail that scattered stations cannot deliver. This dataset is a key part of the contribution, not just a stepping stone to the modeling.

On the technical side, the same data gap makes the modeling problem genuinely different from the one studied elsewhere. Air pollution is strongly correlated in space (neighboring districts share the same weather and pass pollution between them) and strongly correlated in time, since a bad day tends to be followed by another. That makes air quality a natural fit for spatiotemporal graph neural networks, which treat locations as nodes in a graph and pass information across space and time [4], [5]. Almost all earlier ST-GNN work on air quality, however, was done in China, South Korea, and the United States, on dense networks where nearly every node has a sensor and a graph based on distance is a safe default [6], [7]. Bangladesh is the opposite case: most districts have never been measured, and the right graph between them is unknown. A model built for this setting therefore has to do something the earlier work did not need to — generalize to districts it never saw measured, and figure out the connections between districts rather than being handed them. That is the gap the proposed model is designed to fill, and addressing it on a dataset that did not exist before is the technical motivation for this project.

1.3 Objectives

The objectives of this project are enumerated as follows:

1. To construct the first district-level PM2.5 dataset for all 64 districts of Bangladesh by integrating satellite, meteorological, and ground-based data sources.

2. To produce a spatially and temporally complete field through machine-learning calibration while preserving transparent, per-record provenance that distinguishes measurements from predictions.

3. To validate the dataset, and in particular its predictions in unmonitored districts, against an independent satellite-derived PM2.5 reference product.

4. To characterize the dataset through a rigorous exploratory analysis that distinguishes findings grounded in real measurements from those that are artefacts of the calibration model.

5. To establish the first ST-GNN benchmark for the region by implementing and fairly comparing baseline, sequence, established graph, and novel hybrid models under a common evaluation protocol.

6. To design an evaluation methodology appropriate to severe label sparsity, separating performance in well-observed districts, unmonitored (zero-shot) districts, and independent validation.

7. To determine, whether and where spatial graph structure improves prediction relative to purely temporal models.

1.4 Methodology

The work runs in two phases. In the dataset phase, satellite data (Sentinel-5P TROPOMI trace gases and MODIS MAIAC aerosol optical depth) and weather reanalysis (ERA5-Land) are pulled through Google Earth Engine and averaged to daily values around each district headquarters. Ground-truth PM2.5 comes from two networks that complement each other: the OpenAQ record, which goes back to 2019 but is almost all Dhaka, and the Department of Environment network, which covers more districts but only from 2023 and reports Air Quality Index rather than PM2.5 — so its values are converted back to PM2.5 by reversing the US EPA formula. A Random Forest model is then trained to predict PM2.5 from the satellite and weather inputs and used to fill in every district-day that has no measurement. Measured and predicted values are combined into one field, each tagged with its source, and the predictions are checked against an independent reference product.

In the modeling phase, the dataset is set up as a next-day prediction task and used to build and test the proposed model. The proposed model keeps a strong temporal core — a recurrent encoder that reads each district's recent history — and adds a spatial part where a learned self-attention over districts acts as a graph whose connections are learned from data, in place of the fixed distance- or wind-based graphs that established models use. This spatial correction is added through a gate that starts at zero, so the model begins as the plain temporal model and only uses the spatial signal if it actually helps — which means it can never be hurt by a spatial signal that turns out to be weak. To place the proposed model in context, it is compared against a full range of alternatives — simple baselines, a gradient-boosted tabular model, a purely temporal sequence model, and four established ST-GNN architectures on three different fixed graphs — all under a three-track evaluation protocol, with every important comparison repeated across many random seeds and checked for statistical significance.

1.5 Project Outcome

The project produced three main outcomes.

The **headline outcome** is the proposed model. Its learned-attention graph gives a small but statistically real improvement over a strong, tuned sequence model in exactly the case that matters for Bangladesh: the unmonitored districts, where a district has no history of its own and must rely on others. In these zero-shot districts the proposed model reaches a coefficient of determination of about 0.761, against 0.753 for the sequence baseline — a gap that holds up across ten random seeds on both a paired t-test and a Wilcoxon signed-rank test. The improvement is small but solid, and the testing shows it depends on two things: the graph has to be learned, not fixed, and the spatial attention has to be kept simple. Fixed graphs based on distance or wind give no such gain, and adding time-varying complexity to the attention actually takes the gain away.

Just as important is what the rigorous testing ruled out, because those negative results are part of the contribution. No fixed-graph model beat the plain sequence model. A wind-based "transboundary transport" graph looked like a winner in a single run but turned out to be pure noise once it was tested across multiple seeds — a result worth stating plainly, because it shows why the multi-seed discipline was necessary and how easily a false finding could have slipped through. Every model landed in the same narrow zero-shot range of roughly 0.74 to 0.76. That tight grouping indicates a data ceiling, not a model ceiling. This means that the main issue limiting predictions in unmonitored districts is the absence of ground-truth measurements, not the skill of the model. This finding is very helpful for future work because it shows that adding more monitors will be beneficial, while creating fancier models probably will not.

The **second outcome** is the dataset that enables all of this: 163,584 district-day records with documented sources, an honest calibration accuracy (held-out coefficient of determination of about 0.64), and independent agreement of about 0.84 with the reference product, even in districts that have never been monitored. The **third outcome** is the benchmark and evaluation protocol themselves, which is the first for the region and sets a clear, tested standard that any future model for Bangladeshi air quality will need to meet.

1.6 Organization of the Report

The remainder of the report is organized as follows. Chapter 2 provides background on air-quality science, satellite remote sensing, graph neural networks, reviews related work, and ends with a gap analysis that relates this project to it. Chapter 3 describes the methodology used to build the dataset and design the proposed model and the benchmark, including the alternatives considered and the reasons for each choice. The implementation and results are presented in Chapter 4 including the exploratory analysis of the dataset and the full benchmark results with statistical testing. Chapter 5 deals with engineering standards and the impact of the project to society, environment and sustainability and its ethical side. Chapter 6 concludes with a summary of the contributions, limitations and directions for future work.

Chapter 2

Background

This chapter explains the problem this research tackles, reviews what other people have already done, and shows where the gaps are.

2.1 Introduction

Fine particulate matter (PM2.5) is one of the most harmful air pollutants. Even though Bangladesh has some of the worst PM2.5 levels in the world, the country has very few ground AQI monitor stations. Most of its 64 districts have never been measured at all. To study air quality across the whole country, we need a way to fill in the blanks: to estimate daily PM2.5 in places and on days where no monitor exists.

PM2.5 research has two useful tools that rarely get used together. Satellite-plus-machine-learning models can estimate pollution from above, even where there are no ground sensors. Meanwhile, ST-GNNs can learn the spatial and temporal links between one location and the next. This thesis joins the two and applies them to Bangladesh, a setting where neither has really been put to the test.

The rest of the chapter is split into four parts. Section 2.2 covers the existing research in four areas: satellite and machine learning approaches to estimating PM2.5, including the random-forest work done across South Asia; spatiotemporal graph neural networks built for air quality and similar forecasting problems; air quality studies done in Bangladesh, and what they leave out; and methods for making predictions when data is scarce, such as learned graphs and kriging. Section 2.3 explains the main models in plain terms, so the later chapters are easier to follow. Section 2.4 draws the research together to show exactly where the gaps are. Section 2.5 presents the summary.

2.2 Literature Review

2.2.1 Significance of PM2.5 in Bangladesh

PM2.5 (particulate matter) is airborne particles about 2.5 micrometers in diameter. For context, that is roughly thirty times thinner than a human hair. Because they are so small, they travel deep into the lungs and pass into the bloodstream. Exposure to these particles can cause heart disease, stroke, respiratory illness, and early death [8],[1]. According to the WHO, outdoor air pollution caused an estimated 4.2 million premature deaths worldwide in 2019. About 89% of those deaths occurring in low- and middle-income countries [8]. PM2.5 is the air pollutant tied to the largest share of this burden. In 2021 the WHO tightened its air quality guidelines, the recommended 24-hour mean for PM2.5 was set at 15 $\mu g/m^3$ and the annual mean at

5 μg/m³ [2].

Bangladesh sits far above these limits. In the 2024 IQAir World Air Quality Report, Bangladesh was the second most polluted country in the world (behind Chad at 91.8 μg/m³), with an annual average PM2.5 of 78.0 μg/m³ — more than fifteen times the WHO guideline [9]. The country has ranked among the worst in the world every year, consistently. This makes Bangladesh an urgent case for better air quality information. But the monitoring network is sparse and that is what this thesis is about.

2.2.2 Satellite and machine learning PM2.5 estimation

Ground monitors do a good job of measuring PM2.5 accurately, but only at single points. Satellites see the whole country, but they don't measure PM2.5 directly. They measure aerosol optical depth (AOD), which is a measure of how much light aerosols block in a column of air. The basic idea of satellite PM2.5 estimation is to convert AOD (plus weather and land data) to the ground PM2.5. It's a complicated relationship, depending on humidity, the height of the aerosols, particle type, and more, so researchers looked to flexible models.

The global geophysical school (van Donkelaar et al.) The most influential body of work combines satellite AOD with a chemical transport model (GEOS-Chem) and then calibrates the result to ground monitors using geographically weighted regression (GWR). Van Donkelaar et al. [10] pioneered the first global mapping of PM2.5 from satellite AOD. Hammer et al. extended this to a consistent global series of annual PM2.5 from 1998 to 2018 [11]. In that work the raw geophysical estimates agreed with ground monitors at R^2 = 0.81 (slope 0.90), and the GWR calibration raised cross-validated agreement to R^2 = 0.90–0.92 (slope 0.90–0.97). These products from the Atmospheric Composition Analysis Group (ACAG) are widely used as a reference dataset, and this thesis uses the ACAG product as an independent check on its own gap-filled predictions. The key limitation for our purposes is that these global products are annual or monthly, not daily, and they are produced by one global model rather than trained specifically for Bangladesh.

The tree-ensemble school (Wei and colleagues, over China). A second major line of work uses tree-based machine learning to produce daily, high-resolution PM2.5 maps. The most prominent is the work of Jing Wei and colleagues, who developed a "space-time extremely randomized trees" (STET) model. Their 2020 paper produced daily 1-km PM2.5 over China using MODIS MAIAC AOD plus meteorology, land use, elevation, population, and emissions [12]. They reported strong performance — but, crucially, the headline numbers come from two different cross-validation schemes that give different answers. Their out-of-sample (random) cross-validation reached R^2 = 0.89, while their out-of-station (site-based, spatial) cross-validation was lower at R^2 = 0.88, with RMSE around 10.3–10.9 μg/m³. An earlier Wei et al. paper using a space-time random forest reported a national R^2 of about 0.85 [13].

The South Asia / Indo-Gangetic Plain random forest study (Mhawish et al.). Most directly relevant to Bangladesh is the work of Mhawish et al., published in Environmental

Science & Technology in 2020 [14]. They estimated daily 1-km PM2.5 across the Indo-Gangetic Plain — the polluted region that includes northern India and extends toward Bangladesh — by feeding satellite AOD, columnar water vapour, meteorology, and land use into a random forest. They compared the random forest against a linear mixed-effect model and found the random forest clearly better, with a cross-validated explained variance of $R^2 = 0.87$ and a relative prediction error of 24.5%. This study is a close methodological cousin of the calibration step in this thesis: same region, same kind of inputs, same model family.

Other relevant tree-ensemble and fusion studies include random-forest PM2.5 estimation over the United States with cross-validated R^2 around 0.80 by Hu et al. [15], showing how widely this approach has been applied and how performance varies by region and evaluation method.

2.2.3 Spatiotemporal graph neural networks for air quality and forecasting

The second pillar of this thesis is the spatiotemporal graph neural network. The idea is to treat monitoring locations as nodes in a graph, with edges describing how they relate, and to learn both the spatial links between nodes and the time patterns at each node. Most of these models were first built for traffic forecasting, where sensors on a road network produce exactly this kind of data, and were later carried over to air quality.

DCRNN (Li et al., 2018). The Diffusion Convolutional Recurrent Neural Network treats the spread of information across the graph as a diffusion process — a random walk over the network — and combines this with a recurrent encoder-decoder for time [16]. On the two real-world traffic datasets METR-LA and PEMS-BAY it improved on the state of the art by 12–15%. DCRNN is one of the established architectures benchmarked in this thesis.

STGCN (Yu et al., 2018). The Spatio-Temporal Graph Convolutional Network drops recurrence entirely. It stacks graph convolutions (for space) with one-dimensional convolutions (for time), which makes it fast to train with fewer parameters [17]. It is a purely convolutional take on the same problem and is another benchmark architecture here.

Graph WaveNet (Wu et al., 2019). This model is the most important predecessor of the model proposed in this thesis. Earlier models assumed the graph was fixed and known in advance — for traffic, the road map. Graph WaveNet argued that the given graph may be wrong or incomplete, and that the true dependencies should be learned from the data. It introduced a self-adaptive adjacency matrix, learned through node embeddings, that captures hidden spatial links without any prior graph [18]. It pairs this with stacked dilated convolutions to reach long time ranges. The proposed model in this thesis is in this lineage: it also learns the graph from data rather than assuming one, which is exactly what a setting like Bangladesh — where the right graph between districts is unknown — calls for.

GConvGRU / GCRN (Seo et al., 2018). The Graph Convolutional Recurrent Network generalises a recurrent network so that it works on graph-structured data, by putting spectral graph convolutions inside the gates of a recurrent unit [19]. The GConvGRU variant is another benchmark architecture in this thesis.

A3T-GCN (Bai, Zhu et al., 2021). This model adds an attention mechanism on top of a temporal graph convolutional network, so the model can weigh which past time points matter most rather than assuming recent ones always matter more [20]. It is relevant because this thesis also uses attention, though over space (between districts) rather than over time.

PM2.5-GNN (Wang et al., 2020). This is the domain-specific graph model for air quality. Rather than learning the graph blindly, Wang et al. built it from domain knowledge: cities are nodes, and edges encode geographic relationships and wind transport, with meteorology as node and edge features [21]. A graph neural network then models pollution transport between cities while a recurrent network handles diffusion over time, and the model forecasts PM2.5 up to 72 hours ahead. PM2.5-GNN shows the value of a graph built for the physics of air pollution, but it also assumes a reasonably dense network of monitored cities — an assumption that does not hold in Bangladesh.

Across all these models, two foundations recur — the graph convolution and the attention mechanism — which the next section explains.

2.2.4 Air quality modelling in Bangladesh and the regional gap

Air quality research specific to Bangladesh exists, but it is consistently small in scope. Almost all of it is centred on Dhaka or on a handful of city stations, and almost all of it is evaluated with random splits rather than spatial or temporal hold-outs.

Islam et al. (2023) estimated ground-level PM2.5 in Dhaka using best-subset regression and several ML models. These included random subspace, random tree, and reduced-error pruning tree [22]. Their study covered just two stations in the Dhaka area over 2012–2020. The random subspace model performed best, and random cross-validation was used. Shahriar et al. (2020) modelled particulate pollution at four city stations — Dhaka, Chattogram, Rajshahi, and Sylhet — over 2013–2019 using Gaussian process regression, artificial neural networks, random forests, and support vector machines; Gaussian process regression worked best for Dhaka and Rajshahi while the neural network was best for Chattogram and Sylhet, all validated with a random train/test split [23]. A related study by the same group used CatBoost and other models across three city hotspots, again with a random 80/20 split [24]. Islam et al. (2023) in Aerosol and Air Quality Research looked at meteorological influence on PM across eight stations in six cities using non-linear regression, and explicitly concluded that a single model does not transfer well across different spatial scales [25].

The pattern is clear. The largest-scope Bangladesh studies cover at most a handful of cities and a handful of stations. None covers all 64 districts. None provides a daily, district-level, multi-year nationwide PM2.5 dataset. And none applies a spatiotemporal graph neural network to Bangladeshi air quality. Where nationwide district coverage exists at all, it comes only from the global annual or monthly satellite products described earlier, not from a model trained on Bangladeshi data at a daily, district scale. This is the regional gap this thesis fills.

2.2.5 Prediction under sparse data: learned graphs and kriging

The hardest part of this thesis is estimating PM2.5 in districts that have never been monitored — a "zero-shot" prediction at locations with no training labels of their own. Two ideas from the literature speak to this.

The first is learned-graph methods, already introduced through Graph WaveNet [18]. When you do not know which places influence which, you let the model learn the connections from the data. This is the right tool for Bangladesh, where there is no obvious physical graph linking districts and where monitored districts are few.

The second is spatiotemporal kriging with graph networks. Wu et al. (2021) introduced the Inductive Graph Neural Network for Kriging (IGNNK), which is built specifically to recover signals at unsampled locations [26]. The clever part is that it trains by randomly hiding some nodes and learning to reconstruct them, so the trained model can be applied to brand-new sensors it never saw during training — what they call inductive, or being able to create "virtual sensors". This is conceptually close to the zero-shot district problem in this thesis. The difference is that IGNNK reconstructs from other observed nodes at the same time, whereas this thesis estimates from satellite and weather features, with a learned graph helping share information between districts.

2.3 Background on Key Models

This section explains, in plain language, the relevant models we should go through before next chapter.

Random forest. A random forest is an ensemble of decision trees, each of which was trained on a random subset of the data and a random subset of the features. The predictions are combined by taking an average. It handles messy, non-linear relationships well and is not often badly overfitted. We use a random forest for the calibration step—turning satellite and weather inputs into PM2.5—because it is the proven workhorse for this exact task in South Asia, as shown in the literature in Section 2.2.2 [14].

LSTM (Hochreiter and Schmidhuber, 1997). A Long Short-Term Memory network is a recurrent neural network designed to remember information over long stretches of a sequence [27]. Plain recurrent networks “forget” too fast due to vanishing gradients. The LSTM solves this with a memory cell and gates that decide what to keep, what to forget and what to output. The LSTM is the backbone of this thesis dealing with the time dimension, i.e. the day-to-day evolution of PM2.5 at each district, and is also a strong standalone baseline.

Graph convolutional network (GCN; Kipf and Welling, 2017). A GCN is a neural network that works on graphs instead of grids. Each node updates its own representation by mixing in information from its neighbours, layer by layer [28]. Stack a few layers and information spreads across the graph. The GCN is the simple, efficient template that most of the spatiotemporal models in Section 2.2.3 build on. Its weakness is that it treats every neighbour with a fixed weight set by the graph, which leads to the next idea.

Self-attention and GAT (Vaswani et al., 2017; Veličković et al., 2018). Attention is a mechanism that lets a model decide how much each other element should matter when updating a given element, by computing a learned similarity score between them. It is the heart of the Transformer architecture [29]. The Graph Attention Network (GAT) brings this to graphs: instead of fixed neighbour weights, each node learns how much to attend to each neighbour [30]. This matters enormously for this thesis. Because the right graph between Bangladeshi districts is unknown, the proposed model uses a learned self-attention adjacency — it lets the data decide, through attention, how strongly each district should listen to each other district, producing a data-driven, fully-connected weighted graph.

The ST-GNN benchmark architectures. Chapter 3 compares several established spatiotemporal graph neural networks. To recap their roles: DCRNN models space as diffusion and time with a recurrent encoder-decoder [16]; STGCN uses graph and temporal convolutions with no recurrence [17]; GConvGRU puts graph convolutions inside a recurrent unit [19]; and Graph WaveNet learns the graph itself through an adaptive adjacency matrix [18]. The first three assume a fixed, known graph. Graph WaveNet does not — and that is the idea this thesis carries forward.

The proposed model, in brief. The model proposed in this thesis is an LSTM with a learned self-attention adjacency over districts, added through a gated residual connection. In plain terms: the LSTM handles time, the self-attention layer learns a weighted graph linking all districts, and the gate controls how much of that graph-based information is blended in. It is in the lineage of Graph WaveNet's adaptive adjacency [18] and the attention idea of GAT and the Transformer [29], [30], but adapted to a sparse, mostly-unmonitored setting.

2.4 Gap Analysis

Bringing the literature together, this thesis fills four specific gaps —

Gap 1: No nationwide, district-level, multi-year PM2.5 dataset for Bangladesh. As Section 2.2.4 showed, every prior Bangladeshi study is built on a few city stations, and most are about Dhaka [22], [23], [24], [25].Only global annual or monthly satellite products provide nationwide coverage [11]. There hasn’t been a daily dataset available covering all 64 districts across multiple years. This thesis develops the first: 163,584 daily records in 64 districts for 2019-2025 combining satellite trace gases and AOD, reanalysis weather, and ground measurements, with a random-forest calibration to fill the gaps where only 8.45% of values are observed and 41 of 64 districts have never been monitored.

Gap 2: No ST-GNN benchmark for the region. Spatiotemporal graph models in Sec. 2.2.3 have been evaluated on traffic and Chinese air quality [16], [17], [18], [19], [21], but not Bangladesh. However, no benchmark has been published showing how these architectures perform on Bangladeshi air quality, and no shared protocol for comparing them here. This thesis presents the first such benchmark, evaluating baselines, a gradient-boosted tabular model, an LSTM and four well-established ST-GNN architectures, under a single evaluation protocol with multi-seed statistical testing.

Gap 3: Existing ST-GNNs assume dense monitoring and fixed graphs. For example, models like DCRNN, STGCN, and GConvGRU assume that the graph is known in advance and that there are enough monitored nodes for learning [16], [17], [19]. Bangladesh violates both assumptions: most districts are not monitored, and there is no clear physical graph between them. This motivates a learned-adjacency model in the spirit of Graph WaveNet [18] and IGNNK [26] — one that learns the district graph from data rather than assuming it. This is precisely what the proposed self-attention adjacency does.

Gap 4: Zero-shot evaluation in unmonitored districts is rarely discussed. Much of the literature has used random cross-validation that leaks information and inflates scores [12], [22], [23]. Very few studies check what the model does at a place with no labels of its own. This thesis treats that case as a main result, not an afterthought: zero-shot performance in unmonitored districts is reported as its own evaluation track, alongside well-observed districts and an independent check against ACAG.

Taken together, these four gaps define what this thesis sets out to do: build the dataset that does not yet exist, and use it to test whether learning the graph from data helps where the existing models were never designed to work.

2.5 Summary

This chapter laid out the field this thesis builds on. PM2.5 is a serious health threat and Bangladesh is among the worst-affected countries, yet its monitoring network is sparse and most districts have never been measured [8], [2], [9]. Two research communities offer tools: satellite-plus-machine-learning estimation, where random forests over South Asia are the proven approach [11], [14], and spatiotemporal graph neural networks, where the trend has moved from fixed, known graphs toward graphs learned from the data [16], [17], [18], [21]. Prior Bangladeshi work is city-centric, often evaluated with optimistic random splits, and no nationwide district-level dataset or ST-GNN benchmark exists [22], [23], [25]. Chapter 3 turns these gaps into methods.

Chapter 3

Research Methodology

This chapter explains how the work was done, end to end. It covers the overall approach, how the dataset was built and checked, how the data was shaped for modelling, the design of the proposed model, and the protocol used to test it fairly against every alternative.

3.1 Introduction

The work splits into two phases that depend on each other. The first phase builds a dataset, because none existed: a daily, district-level record of PM2.5 for all 64 districts of Bangladesh. The second phase uses that dataset to build and test a spatiotemporal model, and to ask a specific question — does learning the graph between districts from data help in a country where most districts have never been measured?

3.2 Methodology Overview

Figure 3.1 shows the complete pipeline from raw data collection to final results. Reading top to bottom: satellite and weather data is pulled and aggregated to each district and day; a Random Forest learns to turn those inputs into PM2.5 and fills the gaps where no monitor exists, producing the dataset; the dataset is reshaped into a spatiotemporal tensor and three candidate district graphs; a benchmark of models baselines, a tabular model, a sequence model, and several graph models including the proposed one is trained on that tensor; and every model is scored through the same three-track evaluation, with multi-seed significance testing on the comparisons that matter.

Two design principles are woven throughout the whole pipeline and deserve to be stated upfront. First, the division between data and ground truth is maintained honest: all PM2.5 values in the dataset are marked as either measured or predicted, and no accuracy number in this thesis is ever calculated against a predicted value – only against real measurements. Second, there is no leakage over time. The data is split by year – train on 2019 to 2023, validate on 2024, test on 2025 – and every statistic touching the model, from feature standardisation to the calibration itself, is computed from the training years only and then applied forward.

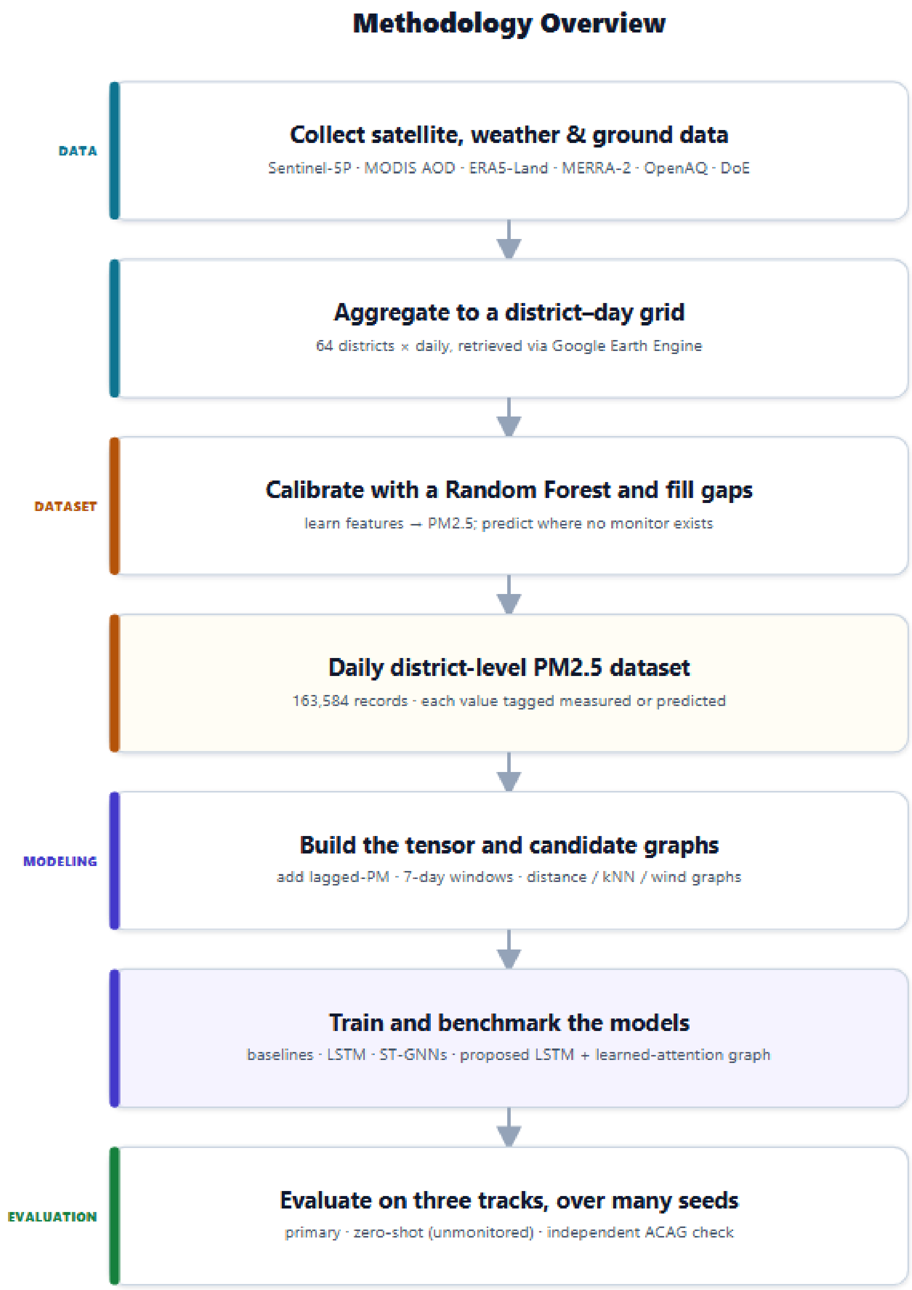

Figure 3.1: Methodology Overview

3.3 Dataset Collection

The goal of this phase is a full grid: 64 districts x 2,556 days, or 163,584 district-day records, each with a PM2.5 value and set of predictor features. The trouble is that almost none of those cells have any real measurement in them. So the dataset is built in three stages – gather the inputs, gather whatever ground truth exists, then learn a model that fills the rest – and then an independent check.

3.3.1 Predictor data from satellite and reanalysis

All predictor data is retrieved via Google Earth Engine [31] and averaged over a fixed radius around each district headquarters, providing one value per district per day. The inputs are in a few groups:

- **Satellite trace gases and aerosols.** Sentinel-5P TROPOMI [32] provides tropospheric NO2, SO2, CO, O3, formaldehyde and a UV aerosol index. MODIS MAIAC [33] offers aerosol optical depth at 550 nm which is the single most important satellite signal for PM2.5: how much light aerosols block in a column of air.

- **Weather.** ERA5-Land reanalysis [34] provides daily temperature (mean, max, min), dewpoint, relative humidity, surface pressure, precipitation, solar radiation, and the two wind components, plus planetary boundary-layer height. Weather matters because it controls how aerosols form, settle, and disperse — the same emissions give very different surface PM2.5 on a still day versus a windy one.

- **Aerosol species and context.** MERRA-2 reanalysis [35] adds surface concentrations of black carbon, organic carbon, sulphate, dust, and sea salt, which describe what the aerosol is made of. MODIS provides a vegetation index (NDVI) [36], and MODIS FIRMS provides fire radiative power [37], both local and a wind-weighted upwind version meant to capture smoke blown in from outside the country.

After collection, a small fraction of feature values are missing where satellite coverage had gaps — for the MERRA-2 species this is about 4.7%. These gaps are filled with the per-district median, falling back to the global median, so that no model receives a missing input. The PM2.5 target is never imputed at this step; its gaps are exactly what the calibration model is built to fill.

3.3.2 Ground Truth from two networks

Real PM2.5 measurements come from two sources that complement each other, neither of which is enough on its own.

The first is **OpenAQ** [38], which gives daily PM2.5 going back to 2019 but is almost entirely Dhaka — deep in time, narrow in space. After processing this contributes about 2,500 station-days.

The second is the **Bangladesh Department of Environment (DoE) CASE network**, which has stations in more districts but only reports an Air Quality Index (AQI) value, not a raw PM2.5 concentration, and mostly from 2023 onward. To use it, the AQI is converted back to PM2.5 by inverting the US EPA piecewise-linear AQI formula — running the official conversion table backwards to recover the concentration that produced each AQI value. This recovers roughly 13,500 station-day measurements across 26 stations. The inversion is exact within each AQI band, but it does inherit the coarseness of the AQI scale, which is a known and documented limitation of this part of the data.

Combined and aggregated to the district-day grid, the two networks give about 13,831 real measurements — which is only **8.45%** of the full grid. Put plainly, fewer than one in eleven cells has a measurement, 41 of the 64 districts have no measurement at all across the entire period, and the ones that do are weighted heavily toward Dhaka. This sparsity is the central problem the rest of the work is built around.

3.3.3 Calibration: filling the grid with a Random Forest

To fill the remaining 91.55% of cells, a Random Forest is trained to map the predictor features to observed PM2.5. A Random Forest is the natural choice here, and not an arbitrary one — it is the proven approach for exactly this satellite-to-PM2.5 task, used widely across South Asia and beyond [12], [14], [15], and it handles the messy, non-linear way that aerosol optical depth, humidity, and boundary-layer height combine to produce surface PM2.5.

The model is trained only on the observed cells from 2019 to 2023, then it was tuned on cells from 2024, and after that it was tested on the held-out 2025 measurements. On that test year it reaches an R^2 of **0.644**, with a mean absolute error of about 23.7 $\mu g/m^3$. Several model families were compared at this step — linear, Random Forest, and gradient boosting — and the Random Forest was selected because it gave the lowest error on the held-out year.

That 0.644 deserves a clear reading, because it is easy to misjudge against the literature. It is a *temporal hold-out* score: the model is tested on a whole year it never saw, which is a strict test and the one that matches how the dataset is actually used — train on the past, predict forward. Much of the published satellite-PM2.5 work reports higher numbers (often R^2 above 0.85), but those come from random cross-validation, where test days sit right next to training days from the same station and the model is partly interpolating [12], [14]. The two numbers are not measuring the same thing. The honest, forward-in-time number for this dataset is 0.644, and it is reported as such.

Once trained, the Random Forest is applied across the full grid. Each cell ends up with a PM2.5 value and a provenance tag: OpenAQ_direct, DoE_inverted, or predicted. This tagging is what keeps the rest of the thesis honest — anyone using the dataset can filter to measurements only, and every evaluation in this work does exactly that.

3.3.4 Independent validation against ACAG

A calibration model can look fine on held-out monitors and still be wrong in the 41 districts that have no monitors at all — and those districts are the whole point. So the gap-filled predictions are checked against a completely independent product: the satellite-derived monthly PM2.5 estimates from the Atmospheric Composition Analysis Group (ACAG) [10], [11], which are produced by a different method (a global chemical-transport model calibrated to monitors), and which were not used anywhere in building this dataset.

Comparing the monthly average of the predictions against ACAG, the agreement is strong, with a correlation of about **0.84** — and, importantly, it stays at that level even when the comparison is restricted to districts that were never monitored. That is the important test. "The predictions in unmonitored districts are not drifting off into nonsense," it says. An independent source sees the same spatial pattern. It doesn't make the predictions perfect, but it provides real, external evidence that the gap-fill is trustworthy where it counts.

3.4 Spatiotemporal Tensor and Graph Construction

The modelling phase needs the dataset in a different shape, a tensor that can be read by a sequence-and-graph model, and candidate graphs, describing how districts might connect.

The tensor. The dataset is converted into a feature tensor of size [2,556 days × 64 districts × 30 features] with two aligned PM2.5 arrays, one with the full gap-filled values, one with observed values only, with a mask indicating which cells are real. Features are standardized using the mean and standard deviation of the training years only, so no information from 2024 or 2025 leaks backward into the scaling.

The lagged-PM channel — the fairness fix. Early in modelling, the neural models performed badly: the LSTM scored only about 0.44 zero-shot, far below the tabular model. The cause was not the architecture. The tabular model had access to recent PM2.5 history through lagged features, while the neural models did not — they were being asked to predict tomorrow's pollution with no idea what today's was. The fix is to add the previous day's PM2.5 as one extra input channel, standardised like the others, so the feature dimension goes from 30 to 31. With that one change the LSTM jumped to about 0.75 zero-shot. This is the single most important methodological correction in the modelling phase, because without it the whole comparison between graph and non-graph models would have been unfair from the start. Every neural model in the benchmark receives this channel.

The windows. The task is framed as next-day prediction. The models read a 7-day window of all 64 districts and predict each district's PM2.5 on the following day. This yields 1,819 training windows, 359 validation windows, and 357 test windows.

The candidate graphs. Three fixed district graphs are built so that the established graph models have something to work with, and so the proposed model has something to be compared against:

- **A distance graph**, where districts are connected with a weight that falls off smoothly with the distance between their centres (a Gaussian kernel on great-circle distance).
- **A k-nearest-neighbour graph**, where each district is linked to its five closest neighbours, made symmetric so links go both ways (about 450 edges).
- **A wind-transport graph**, built from the prevailing winter wind direction, which connects a district to others that sit downwind of it within 300 km. This one encodes the transboundary-pollution hypothesis — the idea that winter wind carries pollution in from the northwest — and exists specifically so that hypothesis could be tested rather than assumed.

3.5 The Proposed Model

The proposed model answers the question that the fixed graphs cannot. Every graph above is a guess about which districts influence which. In a country where 41 districts have never been measured, and where there is no road network or sensor map to lean on, any such guess is shaky. The proposed model removes the guess: instead of being handed a graph, it learns one from the data.

The design keeps the part that already works – a sequence model for time – and layers on top a learned spatial part, switched in gently through a gate. The rest of this section walks through it bit by bit, and then gives the exact forward pass.

3.5.1 The idea

Air pollution in one district is not independent of its neighbours, districts share weather and exchange pollutants. A pure sequence model would treat each district separately and would ignore this. It is used for a graph model. But only if the graph is correct. The idea behind the proposed model is that the model itself can find useful connections between districts, and that these connections need not correspond to geography. Another district, distant but sharing a pollution regime, might be best predicted, and a learned graph can capture that whereas a distance graph cannot.

This is the idea behind adaptive-adjacency models in the larger literature, where the graph is viewed as something to learn rather than something to assume [18]. The novelty here is the setting: applying that idea to a problem of regional air-quality where the ground-truth is very sparse and most nodes of the graph are never observed. That sparsity also explains why the spatial part is added carefully, through a gate, rather than allowed to dominate.

3.5.2 The temporal core

Each district's 7-day window of 31 features is read by a shared Long Short-Term Memory (LSTM) network [27] — shared meaning the same weights process every district, so the model learns one general notion of how pollution evolves in time rather than 64 separate ones. The LSTM has a 64-unit hidden state and two stacked layers with dropout for regularisation. For each district, the hidden state at the final day of the window is taken as that district's temporal embedding: a single 64-number summary of where its pollution has been and where it is heading. Stacking these across districts gives a [64 × 64] matrix — 64 districts, each with a 64-dimensional embedding.

On its own this temporal core is already a strong model — it is the tuned LSTM that serves as the main baseline in the benchmark. The proposed model's job is to see whether a learned spatial layer can add anything on top of it.

3.5.3 The learned attention graph

The spatial layer is a single self-attention operation over the 64 district embeddings [29]. Self-attention lets every district look at every other district and decide, through a learned score, how much each one matters to it.

Concretely, the 64 district embeddings are projected into queries, keys, and values through three small learned linear layers. Each district's query is compared against every district's key with a scaled dot product, and a softmax turns those comparisons into weights that sum to one. The result is a [64 × 64] matrix where entry (i, j) says how much district i attends to district j.

That matrix is the heart of the contribution, and it is worth naming clearly: **it is a learned adjacency matrix.** It is a fully-connected, weighted, data-driven graph over the districts — the same object a fixed graph provides, except discovered from the data instead of fixed by geography or wind. This is the same principle as the adaptive adjacency of Graph WaveNet [18] and the learned neighbour weighting of graph attention networks [30], and it is what makes the proposed model a graph model with a learned graph, rather than a non-graph model or a fixed-graph model. The attention here is single-head, which keeps the learned graph to one interpretable matrix that can be pulled out and analysed directly.

Multiplying this attention matrix by the value projections produces a spatial correction for each district — a 64-number vector that blends in information from the districts the model has chosen to attend to.

3.5.4 The gate

The spatial correction is not simply added to the temporal embedding. It is added through a single learned gate, as embedding + tanh(gate) × correction. The gate is one scalar number, and it is initialised at zero. Because tanh(0) is zero, the model starts life as a pure LSTM with the spatial layer contributing nothing, and it has to actively learn to open the gate if — and only if — the spatial correction helps.

This design is deliberately conservative, and the reason is the sparse data. With so few labels, a spatial layer that is forced on the model could easily hurt by adding noise. The zero-initialised gate means the spatial layer can never make things worse than the LSTM by default; it can only earn its way in. After training, the gate settles at about 0.104 — meaning the spatial correction ends up contributing roughly 10% of the signal. Small, but, as the results show, real.

3.5.5 The full forward pass

Put together, the proposed model runs as follows for one batch:

1. **Input** — a window of shape [batch, 7 days, 64 districts, 31 features], the 31st feature being the standardised previous-day PM2.5.

2. **Temporal encoding** — reshape to process every district through the shared LSTM, and take each district's final-day hidden state. Result: a [batch, 64, 64] tensor — one 64-dimensional temporal embedding per district.

3. **Learned graph** — project the embeddings to queries, keys, and values; form the scaled dot-product attention matrix [batch, 64, 64], which is the learned adjacency; multiply it by the values to get the spatial correction [batch, 64, 64].

4. **Gated fusion** — combine as embedding + tanh(gate) × correction, so the spatial signal is blended in only as much as the learned gate allows.

5. **Prediction** — a single linear layer maps each district's fused 64-number embedding to one number: its predicted PM2.5 for the next day. Output shape [batch, 64].

The model is trained to minimise error against observed PM2.5 only, so it is never rewarded for matching the Random Forest's own gap-filled guesses.

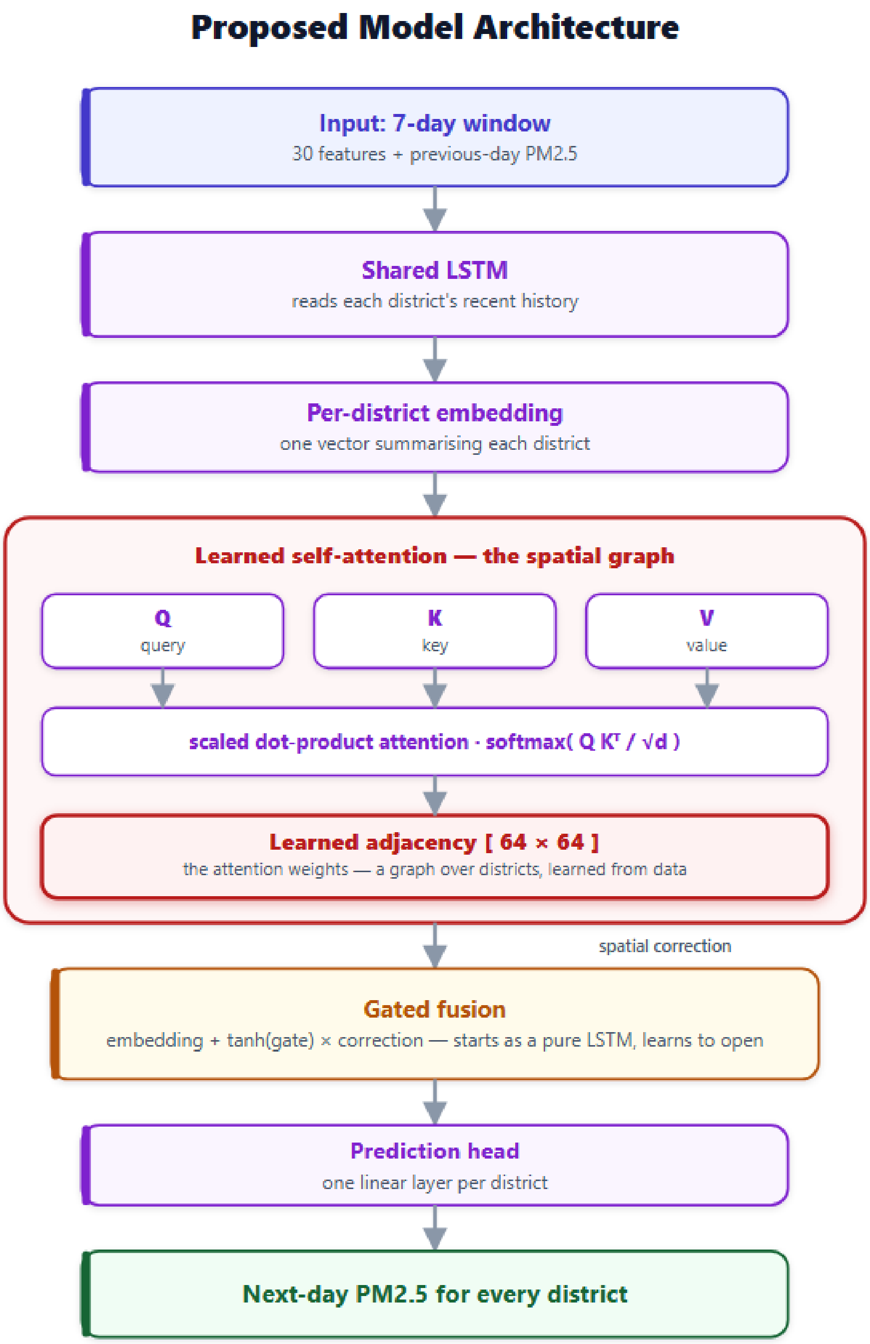

Figure 3.2: Model Architecture

3.6 Benchmark Models and Evaluation Protocol

A single model in isolation says little. The proposed model only means something next to a fair field of alternatives, scored the same way. This section describes that field and the protocol.

3.6.1 The benchmark models

The models are arranged as a ladder, from simplest to most complex, so that each step answers a clear question:

- **Naive baselines** — persistence (tomorrow equals today) and climatology — set the floor that any real model must beat.

- **A tabular model** — gradient-boosted trees with lagged-PM features — represents the strong non-sequence, non-graph approach.

- **A sequence model** — the tuned LSTM described in 3.5.2 [27] — is the strongest non-graph model and the main thing the proposed model must beat.

- **Established ST-GNNs** — STGCN [17], DCRNN [16], GConvGRU [19], and A3T-GCN [20] — are the standard spatiotemporal graph architectures from the literature, each run on the fixed candidate graphs. These test whether an off-the-shelf graph model, with a sensible fixed graph, can beat the sequence model.

- **The adaptations** — the proposed learned-attention model, plus two controlled variants used as ablations: one that adds a *fixed*-graph correction through the same gated residual (to isolate whether the benefit comes from having a graph at all, or specifically from *learning* it), and one that recomputes the attention at every timestep instead of once (to test whether a time-varying graph helps). These last two exist to localise exactly where the proposed model's behaviour comes from.

3.6.2 The three-track evaluation protocol

Standard practice would report one accuracy number on a random test split. That would be misleading here, because a random split mixes well-monitored Dhaka with never-monitored districts and hides the thing that actually matters — how the model does where there is no ground truth. So performance is reported on three separate tracks, and all accuracy is computed against observed measurements only.

- **Track 1 — primary districts.** Forecasting accuracy in the 12 districts that are observed in both the training years and the test year. This is the "easy" case: the model has history for these places. It measures ordinary forecasting skill.

- **Track 1b — zero-shot districts.** Accuracy in districts that are *not* in the training set, measured on whatever test-year observations exist there. This is the hard case and the one the whole thesis is about: predicting pollution in a district the model never learned from. This is where the proposed model is meant to earn its keep.

- **Track 2 — independent field agreement.** Monthly agreement with the ACAG product [10], [11], split out for observed versus never-observed districts, as an external sanity check on the spatial pattern.

- **Track 3 — by monitoring density.** Accuracy split by how well-observed a district is (well-observed versus sparse), to show how performance scales with the amount of ground truth.

3.6.3 Multi-seed significance testing

Neural models give slightly different results every time they are trained, because of random initialisation. A difference between two models can easily be nothing more than a lucky seed. This is not a hypothetical concern in this work — at one point a wind-transport graph appeared to beat the distance graph by a clear margin in a single run, which looked like a real regional-physics finding, and it vanished completely once the comparison was repeated across several seeds. It was noise.

To guard against this, every comparison that matters is run across multiple random seeds — five to ten depending on the comparison — and tested for significance with both a paired t-test and a Wilcoxon signed-rank test. A difference is only treated as real if both tests agree at $p < 0.05$. This rule is strict on purpose, and it is what separates the claims in this thesis that are reported as findings from the ones that are reported as noise.

The results of running this protocol across the full benchmark are presented in Chapter 4.

3.7 Task Allocation

This table depicts the timeline of the principal activities in each period of the project, from week 12 to week 48.

Table 3.1: Task Allocation

Tasks	**Weeks**																		
	12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
Data collection phase																			
Preprocess all the data																			
Model training																			
Interpreting results and paper writing																			

3.8 Project Plan

The work was carried out in the two phases described in this chapter. The first phase, dataset construction, covered collecting the satellite, meteorological, and ground-truth data through Google Earth Engine, aggregating it to the district–day grid, training the Random Forest calibration, gap-filling the full record, and validating it against ACAG. The second phase, modelling, covered building the spatiotemporal tensor and the candidate graphs, implementing the baselines and established ST-GNNs, developing the proposed model, and running the three-track evaluation with multi-seed testing. Writing and figure preparation ran alongside the later modelling work rather than waiting until the end, so results were documented as they were produced. The week-by-week task allocation is given in the table below.

3.9 Summary

This chapter set out how the work was done from end to end. It described the two-phase methodology: building the dataset by fusing satellite and reanalysis predictors with the limited ground truth and filling the gaps with a Random Forest calibration, then validating that calibration against an independent product. It then explained how the dataset is reshaped into a spatiotemporal tensor and candidate graphs, with the lagged-PM channel as the key fairness fix, and set out the proposed model in full — an LSTM temporal core with a learned self-attention adjacency, blended through a zero-initialised gate. Finally, it laid out the benchmark of models the proposed method is compared against and the three-track evaluation protocol, with multi-seed significance testing, that keeps the comparison honest. The results of running this methodology are presented in Chapter 4.

Chapter 4

Implementation and Results

This chapter reports what the work produced. It covers the implementation setup, what the dataset looks like once it is built, the exploratory findings that describe Bangladesh's air pollution and shape the modelling, the calibration results and their independent validation, and finally the full model benchmark, the proposed model, and a look inside its learned graph.

4.1 Introduction

The methodology in Chapter 3 was carried out end to end, and this chapter presents the results at each stage. It begins with the practical setup, then turns to the dataset — first its coverage and honesty, then a set of exploratory findings that describe how pollution behaves across Bangladesh in time and space and what drives it. These findings are not just description; several of them directly justify choices made in the modelling, and they are reported here as results in their own right. The chapter then covers how well the calibration performed and how it was independently checked, the full benchmark of models, the proposed model and its ablations, and what the learned graph turned out to contain. Throughout, the evaluation is reported honestly, including where results are modest, where they are negative, and where the data itself sets a limit.

4.2 Environment Setup

The entire project was implemented in Python, across two computing environments suited to its two phases. Data collection ran on Google Earth Engine, the cloud platform that hosts the satellite and reanalysis archives, so the raw imagery never had to be downloaded — only the aggregated district-day values were exported. The calibration and modelling ran in cloud notebooks (Google Colab) on a single NVIDIA GPU, which was enough for models of this size.

The software stack is built entirely from open-source libraries. Data handling and aggregation used pandas and NumPy. The Random Forest calibration used scikit-learn. The neural models were built in PyTorch, with the established spatiotemporal graph baselines (DCRNN, STGCN, GConvGRU, A3T-GCN) drawing on the PyTorch Geometric Temporal library. The statistical significance tests used SciPy, and the figures were produced with Matplotlib. Table 4.1 summarises the environment.

Table 4.1 — Implementation environment.

Component	Tool
Language	Python
Data collection	Google Earth Engine
Compute	Google Colab, single NVIDIA GPU (CUDA)
Data handling	pandas, NumPy
Calibration model	scikit-learn (Random Forest)
Deep-learning framework	PyTorch
Graph models	PyTorch Geometric Temporal
Statistics	SciPy (paired t-test, Wilcoxon)
Figures	Matplotlib

Before modelling, the dataset is loaded into a single spatiotemporal tensor of shape [2,556 days × 64 districts × 31 features] — the 30 predictors plus the lagged-PM channel — together with aligned arrays of observed PM2.5 and a mask marking which cells are real. Features are standardised using training-year statistics only. The models read 7-day windows of all 64 districts and predict the next day.

The calibration model is a Random Forest of several hundred trees, trained on the observed cells from 2019–2023. The neural models share a common training setup, summarised in Table 4.2. They are trained with the Adam optimiser at a learning rate of 1×10^{-3} and a small weight decay of 1×10^{-5}, with the learning rate halved automatically when validation loss stops improving. The loss is a masked mean squared error computed only on observed cells, so the models are never trained against the Random Forest's own gap-filled values. Training runs for up to 40–50 epochs with early stopping (patience of 8 epochs) on the 2024 validation year. The proposed model and its sequence baseline use a 64-unit, two-layer LSTM core with dropout of 0.2.

Table 4.2 — Neural-model training configuration.

Setting	Value
Optimiser	Adam (lr 1×10^{-3}, weight decay 1×10^{-5})
Learning-rate schedule	ReduceLROnPlateau (factor 0.5, patience 3)
Loss	Masked MSE (observed cells only)
Max epochs / early stopping	40–50 / patience 8 (on 2024)
Batch size	16–32
LSTM hidden size / layers / dropout	64 / 2 / 0.2
Input window → horizon	7 days → next day

Two practices run through the whole setup and are worth stating once. First, the split is strictly temporal — train on 2019–2023, validate on 2024, test on 2025 — and every quantity the models depend on is computed on the training years alone and applied forward, so the future never leaks into the past. Second, every comparison that carries weight is repeated across multiple random seeds (the headline proposed-model-versus-LSTM comparison uses ten seeds, 0 through 9) and tested for significance with both a paired t-test and a Wilcoxon signed-rank test, as described in Chapter 3.

4.3 Exploratory Analysis of the Dataset

Before any modelling, the dataset was explored in depth to understand how pollution behaves in Bangladesh. The findings below describe the seasonal cycle, the spatial geography, the drivers, the spatial coupling between districts, the temporal memory, and the public-health burden. Several of them feed directly into the modelling choices.

4.3.1 The seasonal cycle and the winter trap

Pollution in Bangladesh follows a sharp and consistent yearly rhythm. Winter pollution runs about 4.6 times higher than monsoon pollution, and the cycle is so dominant that the time of year alone explains much of the variation. The monsoon months bring the only relief, yet even then the air is not clean: the WHO 24-hour guideline of 15 μg/m³ is exceeded in every month of the year, including the wettest ones.

The mechanism behind the winter peak is the classic "winter trap" of South Asian air pollution, and the data shows it clearly. Looking at how monthly pollution co-varies with the weather, precipitation (correlation −0.94), temperature (−0.91), relative humidity (−0.85), wind speed (−0.75), and boundary-layer height (−0.54) all move opposite to pollution, while black carbon moves almost perfectly with it (+0.97). In winter, a low boundary layer, weak winds, and no rain trap combustion emissions near the surface and let them accumulate; in the monsoon, rain washes particles out, deep mixing dilutes them, and stronger winds disperse them. The causal levers are precipitation, boundary-layer height, and wind — black carbon's near-perfect correlation reflects that it shares the same trapping cycle as the pollution it is part of, so it is described here as co-varying with PM2.5 rather than driving it.

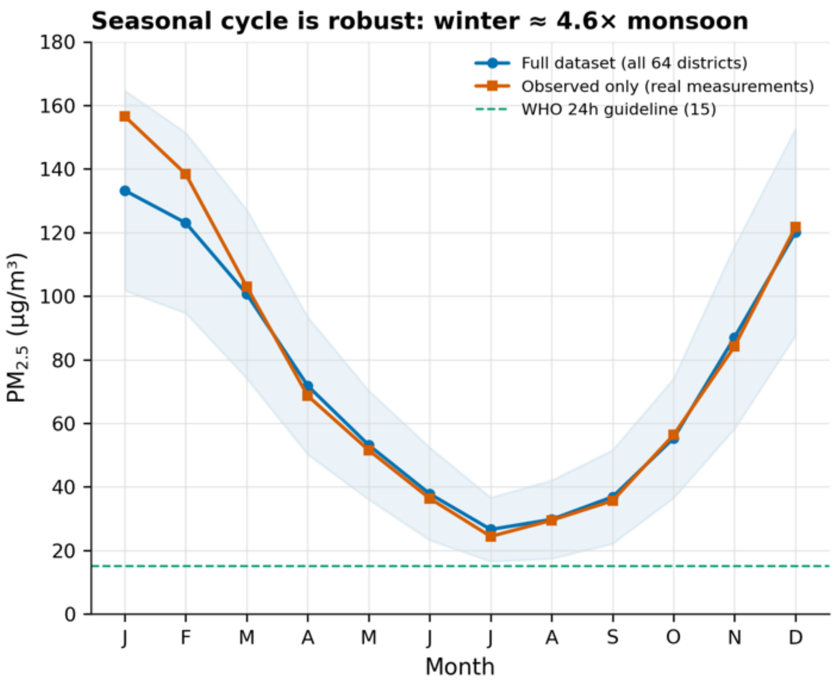

Fig 4.1: Seasonal Cycle

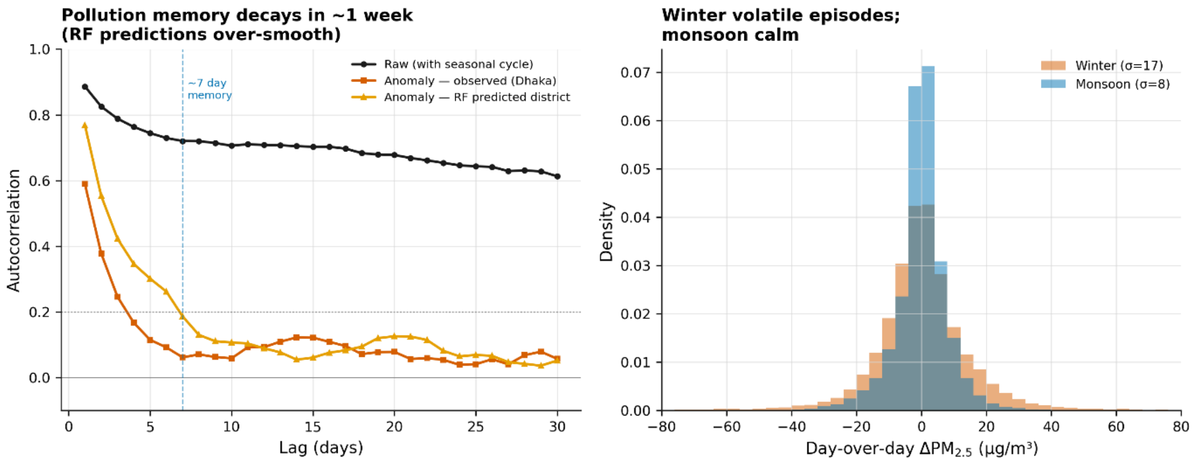

Fig 4.2: Winter Trap Analysis

4.3.2 The spatial geography: where the pollution is

Dhaka-centric is the usual way people think about air pollution in Bangladesh, but a popular study points to a "Northwest pollution belt" of seven districts that is on par with the capital. Neither survives close examination of the data, and what replaces them is wider and better supported.

Even ground truth contradicts the Dhaka-centric view. In the northwest, Rangpur has mean observed PM2.5 of 93.8 μg/m³ — higher than Dhaka's 90.2 μg/m³. Measured directly, the district outside the capital is dirtier than the capital itself. But the seven-district "belt" as a coherent structure does not hold up, for an honest reason: six of those seven districts have never been measured, and their elevated values are pure model predictions. A tree model trained on geographic features will naturally produce smooth spatial patterns in regions with no ground-truth constraint, which can manufacture the appearance of a tidy belt that the data cannot confirm.

What the data does support is a broader, divisional-scale claim. Grouping by division and looking at winter intensity gives a clear and defensible picture:

Table 4.3: Winter Pollution Across Districts

Division	Winter mean (μg/m³)	Monsoon mean
Dhaka	143	30
Rangpur	141	41
Rajshahi	140	36
Mymensingh	134	37
Khulna	123	30
Barishal	107	24
Chittagong	104	25
Sylhet	97	32

The four most polluted divisions in winter are all in the north or northwest, and the south and east are markedly cleaner. This is the defensible spatial claim of the thesis: **northern Bangladesh is the winter pollution core**, supported by both ground truth (Dhaka, Rangpur) and, as shown below, by independent satellite data across the whole region.

4.3.3 What drives the pollution

Bangladesh's PM2.5 is combustion-dominated, and the data says so consistently. Among all the available drivers, the strongest correlates with observed pollution are combustion-related: black carbon (correlation +0.74), sulfate (+0.67), and organic carbon (+0.63). Carbon monoxide (a direct combustion tracer) is the best single predictor of trace gases from the Sentinel-5P satellite (+0.56 overall, and +0.78 at the monthly scale), ahead of nitrogen dioxide (+0.40) and aerosol optical depth (+0.39). Desert dust, on the other hand, is essentially irrelevant (correlation near zero), which confirms the fact that this is a combustion problem, not a wind-blown-dust one.

One striking number emerged from a feature-importance check and is worth careful explanation: black carbon alone was assigned more than half of the importance by the model. That looked worrying, like a leak, so it was checked. It is not leakage. Black carbon is a physical component of fine particulate matter, so we would expect a strong correlation. Removing it does not affect calibration accuracy because the model simply reroutes through correlated features. Black carbon alone attains a R^2 of ~0.47, a large portion of the full model's skill. The honest reading is that its high importance reflects redundancy among correlated combustion signals. The substantive finding underneath stands: combustion aerosols and tracers carry the signal.

The dataset also offers quantitative support for the transboundary-transport hypothesis — the idea that winter winds carry pollution in from crop-residue burning in northwest India. A purpose-built feature measuring upwind fire activity turns out to be about four times as important to the model as local fire activity. The effect is modest in absolute terms but clear in direction, and it is the directional signature the hypothesis predicts. It is also a good example of why both linear and rank correlations were always reported: upwind fire has a near-zero linear correlation with pollution (0.04) but a strong rank correlation (0.39), because the signal lives almost entirely in the long tail of high-burning days. A linear-only analysis would have missed it.

4.3.4 Spatial coupling and the choice of graph

For the modelling, the key question is whether districts are coupled in space in a way a graph can exploit. They are, but with an important nuance. Pairs of districts that are close together do correlate more strongly than distant ones — about 0.80 for districts within 100 km, falling toward 0.70 beyond 350 km. So there is real, distance-dependent spatial structure.

But the curve falls to a high floor, not to zero, and that floor is the finding. Most of the co-movement between districts is not local coupling at all; it is the shared national seasonal cycle that every district experiences together. The genuinely distance-dependent part — the part a spatial graph can add over a purely temporal model — is real but secondary. The practical consequence is that the graph between districts should be **sparse**, not dense: a dense graph mostly re-encodes the shared seasonal signal that the temporal model already captures, and adds noise on top. This was tested directly by scoring three candidate graphs against the observed correlation structure — a dense distance graph scored 0.15, a sparse five-nearest-neighbour graph scored 0.24 (the best), and a wind-aware directional graph scored 0.13. The sparse graph won, which is why it was the primary fixed graph carried into the benchmark.

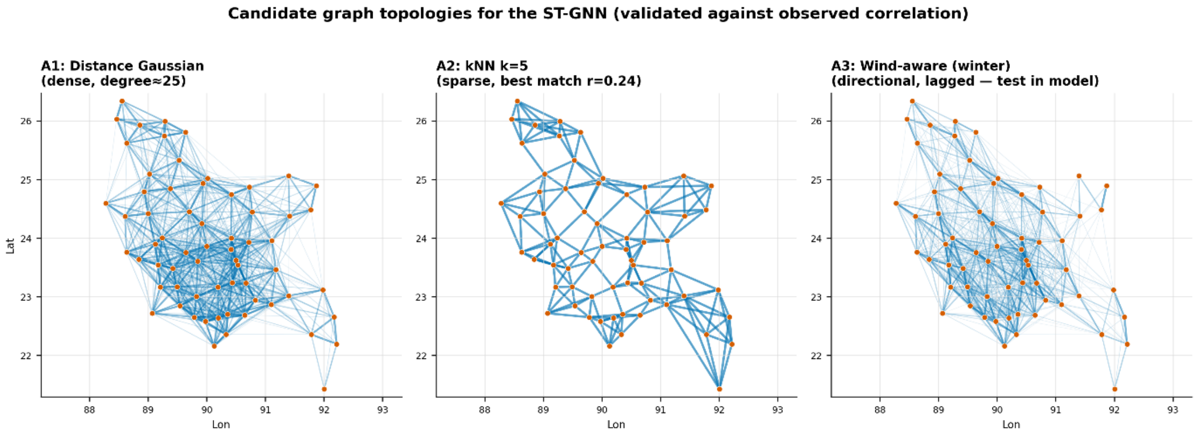

Fig 4.3: Choice of Graph

4.3.5 How far back the memory goes

The other question the modelling needed answered was how much history is useful. After removing the seasonal cycle, the day-to-day "memory" of pollution decays quickly: the correlation of today's anomaly with one a day earlier is about 0.59, falls to 0.25 by three days, and reaches roughly 0.06 — effectively nothing — by seven days. This directly justifies the 7-day input window used by every model in Chapter 3.

This analysis also identified a real limitation of the calibration, by comparing the memory of an observed district against a predicted one. The anomalies predicted for the district decay more slowly than those in the real district – the model creates a temporal smoothness that the real air does not possess. So the gap-filled series is smoother than reality, which is worth remembering if you want to capture short, sharp episodes. It is also an argument for spatio-temporal modelling: a good model should reproduce realistic short-term variability rather than oversmooth it. This is consistent with the fact that day-to-day fluctuations are about twice as large in winter as in monsoon, and that the severe single-day spikes which cause acute health effects are almost exclusively a cold-season occurrence.

4.3.6 The public-health burden

The exploratory analysis closes on the human scale of the problem, and the numbers are stark. The average district breaches the WHO guideline of 15 μg/m³ for a 24-hour period on some 356 days each year, leaving just about nine days of clean air per annum. 57 of the 64 districts have less than 30 clean days per year and in the worst districts the guideline is basically never met for the whole seven year record . It is not a Dhaka problem but a national problem.

The burden is also very seasonal. The only time there's clean air is in a thin sliver around July and August. December and January are characterised by air in the "unhealthy" range or worse. One caveat worth noting for the most extreme categories: The calibration cannot produce values above its empirical ceiling of about 256 μg/m³, so the full dataset undercounts the very worst days, and the observed records are the trustworthy source for the severe-episode burden, which show a meaningfully larger share of “very unhealthy” days.

4.4 Calibration Results and Independent Validation

The Random Forest calibration is the model that fills the 91.55% of the grid with no measurement. On the held-out test year, 2025, it reaches an R^2 of 0.644 with a mean absolute error of about 23.7 μg/m³, and it was chosen over gradient-boosted and linear alternatives because it gave the lowest error on that year. As discussed in Chapter 3, this should be read as a strict, forward-in-time score, not as underperformance against the published satellite-PM2.5 literature, whose higher numbers come from random cross-validation that lets the model interpolate between days it has already seen.

One ablation result is worth reporting on its own. Adding the engineered features — boundary-layer height, fire, vegetation index, and the MERRA-2 species — on top of the satellite-and-weather baseline barely moved accuracy, improving R^2 by only about 0.01. This is a genuine, useful finding: the baseline already captures most of the *learnable* signal at this resolution, and the remaining ceiling is set by how little ground truth exists, not by missing features. The extra features still earn their place in the explanation of the pollution, even where they do not improve point prediction.

The more demanding test is whether the predictions hold up in the 41 districts with no monitor, and for that the gap-filled predictions were compared against the independent ACAG satellite product, which played no part in building the dataset. Across 3,072 overlapping district-months the agreement is strong, and — this is the key result — it is just as strong in the never-observed districts as in the observed ones:

Table 4.4: Independent Validation Results

Districts	n	Correlation with ACAG
With ground truth	1,104	0.83
Never observed	1,968	0.84
Overall	3,072	0.83

An independent, authoritative source, built by a completely different method, sees the same spatial pattern in exactly the districts where this dataset has no ground truth of its own. That is the single strongest piece of evidence that the gap-fill is trustworthy where it cannot otherwise be checked, and it independently corroborates the northern-pollution-core finding from earlier in the chapter.

One honest nuance: the disagreement with ACAG is spatially structured rather than random. This dataset runs somewhat high relative to ACAG in the north and northwest and somewhat low on the coast and southeast. The two sources agree that the northwest is elevated, but differ on exactly how extreme it is, so the fair synthesis is that the northern pollution is real and confirmed, with its precise magnitude likely sitting between the two estimates.

4.5 Benchmark Results

All models were evaluated on the same two main tracks: forecasting accuracy in the 12 core districts that have a history in both training and test years, and zero-shot accuracy in districts the model never learned from. For Bangladesh, it's the zero-shot track that matters, since most of the country is unmonitored. The results are shown in Table 4.3.

Zero-shot is the accuracy in unmonitored districts. The LSTM and the proposed model are averaged over 10 seeds; other graph models are representative runs from the survey.

Table 4.5 — Benchmark results (R^2).

Model	Primary R^2	Zero-shot R^2
Gradient boosting (tabular + lags)	0.826	0.736
STGCN (fixed kNN graph)	0.740	0.610
A3T-GCN (fixed kNN graph)	0.740	0.604
DCRNN (fixed kNN graph)	0.804	0.709
GConvGRU (fixed kNN graph)	0.815	0.725
GConvGRU (fixed wind graph)	0.818	0.741
Dynamic-attention variant	0.810	0.738
LSTM (tuned sequence model)	0.830	0.754
Proposed model (learned attention)	**0.830**	**0.761**

Two things stand out. First, on the zero-shot track, no off-the-shelf graph model beat the simple LSTM. The best of them, GConvGRU on the wind graph, reached 0.741, still below the 0.754 of the LSTM. The purely convolutional graph models, STGCN and A3T-GCN, performed significantly worse at around 0.61, because they lack the proper sequence modelling that the recurrent and sequence models have. Adding a fixed graph didn't help, in fact it often hurt.

Second, more telling, the best models are all in a narrow range. When a model has a real temporal core, the LSTM, the recurrent GConvGRU, the proposed model, its zero-shot accuracy is around 0.74-0.76 and nothing goes beyond about 0.76. This is the central empirical finding of the thesis. It is a ceiling, not a cluster of every model: the weaker architectures sit well below it. But the fact that very different strong models all converge to the same upper limit, and that even a learned-graph model only nudges it, points to the conclusion that the binding constraint is the amount of ground-truth data, not the capacity of the model. This echoes the calibration ablation in §4.5, where extra features also failed to lift the ceiling. More monitoring would raise it; a fancier model will not.

4.6 The Proposed Model and Its Ablations

The proposed model was the only one that beat the plain LSTM in the districts it had never seen during training. The improvement is small, but it holds up when you test it properly. Across ten runs with different random starts, its zero-shot R^2 averaged 0.761 against the LSTM's 0.754, and it came out ahead in eight of those ten runs. Both significance tests agreed the difference is real at the 0.05 level.

Table 4.6 — Proposed model vs. LSTM (zero-shot districts)

Model	Zero-shot R^2	Wins	Significance
Proposed (learned attention)	0.761	8 / 10 seeds	t-test p = 0.037 · Wilcoxon p = 0.049
LSTM (baseline)	0.754	—	reference

We want to be honest about the size of this. It's a small gain, and the Wilcoxon p-value of 0.049 sits right on the line. What makes it a real finding rather than luck is that it's consistent across seeds, it passes both tests, and it shows up exactly where the model was designed to help — in the unmonitored districts. On the monitored districts, where the LSTM already has history to lean on, the two models tie (0.830 each), which is what we'd expect: the spatial help only matters when a district has no data of its own.

That naturally raises a question — *how do we know the design choices behind the model actually matter?* To answer it, we ran two ablations. An ablation just means breaking one part of the model on purpose and seeing whether performance drops, which tells you whether that part was pulling its weight.

Our model makes two specific choices: the graph is **learned** from data, and the attention is computed **once and kept static**. So we tested each choice by breaking it.

Table 4.7 — Ablations: does each design choice matter?

Variant	**Zero-shot R^2**	**What it tells us**
Proposed (learned + static)	0.761	the full design
Fixed graph instead of learned	0.750	ties the LSTM → the benefit needs a *learned* graph, not just any graph
Dynamic attention instead of static	0.738	worse → a time-varying graph over-complicates it; *static* is better

The first ablation swaps the learned graph for a fixed one, fed in the same way. It simply falls back to the LSTM's level — the model's gate learned to ignore the fixed graph entirely. So having a graph isn't the point; having a *learned* one is. The second ablation recomputes the attention at every timestep instead of once. That made things worse, because with so little data the extra flexibility just adds noise. Together, the two results show the design isn't arbitrary: the graph has to be learned, and the attention has to stay simple.

There's one last result worth including, because it's the clearest reason to trust everything above. Early on, the wind-transport graph looked like it beat the plain distance graph by a comfortable margin — about +0.016 in a single run — which was exciting, because it looked like real evidence of pollution blowing in across the border. But when we repeated it across several random seeds, that margin collapsed to about +0.001. It was never a real effect; it was just the noise of a single lucky run.

Table 4.8 — Why we test across seeds: the wind-graph "result"

Wind graph vs. distance graph	R^2 gain	Verdict
Single run	+0.016	looked like a real transboundary effect
Averaged over seeds	+0.001	seed noise — dropped

We report this deliberately. Catching a false result like this is exactly why every headline comparison in this work is run over many seeds and checked with two significance tests — and it's what lets us stand behind the small gain we do claim.

4.7 Summary of Results

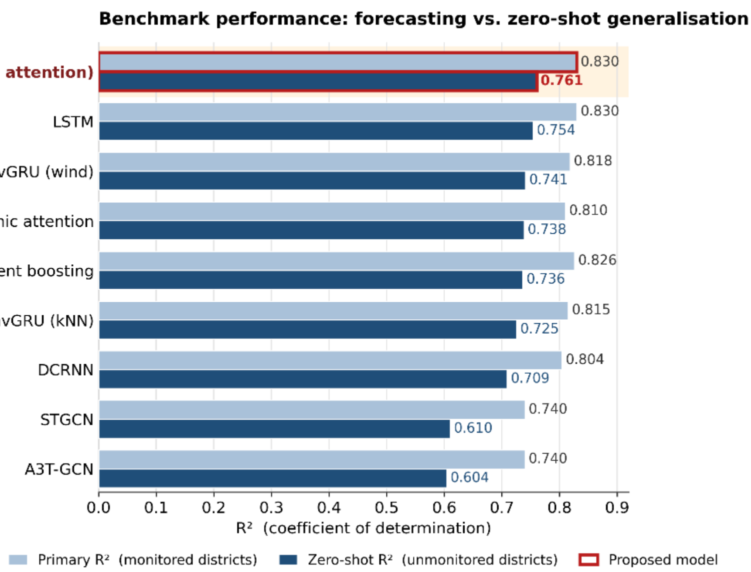

Fig 4.4: Results Analysis of all Models

The results support a clear and honest story. The dataset is structurally complete, but its ground truth is sparse and Dhaka-biased, with 41 districts never measured — which is why provenance and independent validation matter so much. The exploratory analysis revealed a pollution problem that is severe, national, and seasonal: a winter trap that pushes concentrations to several times the monsoon level; a combustion signature dominated by black carbon and carbon monoxide; a northern pollution core that replaces the older Dhaka-centric and “northwest belt” stories; real but secondary spatial coupling that argues for a sparse graph; a short seven-day memory that sets the modelling window; and a public health burden of roughly 356 exceedance days a year. The calibration scores 0.644 on a strict forward hold-out and agrees with the independent ACAG product at 0.83–0.84 even where there are no monitors. In the benchmark, no fixed-graph model beat a well-tuned LSTM, and the strongest models across families converged to a zero-shot ceiling near 0.76. The proposed learned-attention model is the only one to cross the LSTM, by a small but statistically real margin in the unmonitored districts, and only when its graph is learned and its attention kept static. Inside, that learned graph is a mild proximity-based smoothing rather than a transport map. And the ceiling — seen in both the feature ablation and the model benchmark — is the most useful result of all: for nationwide PM2.5 in a country like Bangladesh, the path forward is more ground monitoring, not heavier models. Chapter 5 turns to the standards, impact, and engineering context of this work, and Chapter 6 concludes.

Chapter 5

Engineering Standards and Design Challenges

This chapter places the project in its wider engineering context. It covers the standards the work followed, its impact on society and the environment, the ethical questions it raises, how it maps onto the attributes of a complex engineering problem, and how it was managed and resourced.

5.1 Compliance with the Standards

5.1.1 Software Standards

The project follows standards for scientific software and data. All data are stored in open, non-proprietary formats—comma-separated values for the published dataset and the compressed NumPy archive format for the modeling tensor package—ensuring that the data are accessible without specialized or licensed software. The code is coded in Python using open source libraries with permissive licenses such that the entire pipeline can be replicated at no cost. Geospatial data are in standard conventions for coordinate reference (decimal degrees of latitude and longitude) and in physical units with all quantities in International System units and clearly documented in the data dictionary. The design of the dataset follows the FAIR principles for scientific data (findability, accessibility, interoperability and reusability) by means of its open format, complete documentation, transparent provenance flags and intended release under an open licence with citation of the underlying data providers.

Reproducibility, which is itself an emerging standard of good practice in computational research, is treated as a first-class concern. The modeling tensor package is generated by a self-verifying routine that audits its own output against the source data; random seeds are controlled and reported; the temporal data split is fixed and documented; and all stochastic comparisons are repeated across multiple seeds with statistical testing. These measures align the work with contemporary expectations for reproducible machine-learning research.

5.1.2 Methodological standards for satellite-derived data

The satellite products are themselves the output of standardized, peer-reviewed retrieval algorithms: the MAIAC algorithm for aerosol optical depth, the operational TROPOMI retrievals for trace gases, and the ERA5-Land reanalysis. Each of these has documented quality conventions that the project respects, for example in how aerosol-optical-depth quality flags are handled, and in the treatment of missing retrievals as genuine gaps rather than zeroes.

5.2 Impact on Society, Environment and Sustainability

5.2.1 Impact on Life

The most obvious human impact of the project is that it can contribute to the evidence base for air quality management in a country where air pollution is a leading environmental health risk. A validated, publicly available daily record of PM2.5 at the district level allows exposure assessment at a level of detail that informs epidemiological research, individual and institutional decisions about protective behaviour during high pollution periods and underpins targeting of interventions to the most affected populations.

5.2.2 Impact on Society & Environment

The data and analysis can offer the public and policy a new lens on the problem, making it clear that pollution in Bangladesh is a national issue, not confined to Dhaka, and that the average district exceeds international guidelines for much of the year on the great majority of days. The identification of a combustion-driven winter cycle and the validation of the transboundary impact on the prevailing winter winds are relevant for domestic emissions policy and for regional cooperation on air quality. The work is observational and analytical from an environmental perspective and requires no direct environmental cost beyond modest computation.

5.2.3 Ethical Aspects

There were several ethical considerations. The most important thing is to be honest about uncertainty: since much of the dataset is model-imputed rather than measured, the project is careful not to present predictions as measurements, puts provenance flags on every record, restricts all analysis and evaluation to observed data, and tests predictions in unmonitored districts against an independent source. To do otherwise—to allow imputed values to be mistaken for ground truth—would risk misleading downstream users and policymakers, and the avoidance of that risk is treated as an ethical obligation. A second consideration is the responsible release of data and code. Credentials or private identifiers used in the collection of data are scrubbed before release, and only openly licensable, non-personal environmental data are published. A third is the avoidance of overclaiming: the modelling contribution is presented as a rigorous benchmark with a small, honestly-characterized positive result, rather than inflated into a claim of a transformative new method, in keeping with the scientific responsibility to represent results faithfully.

5.2.4 Sustainability Plan

The dataset is designed for sustainability and reuse. Because it is built entirely from freely available, continually updated data products through a documented and reproducible pipeline, it can be extended forward in time as new satellite, reanalysis, and monitoring data become available, without redesign. The open format and documentation are intended to allow other researchers to maintain, extend, and build upon the dataset, so that it can serve as durable infrastructure for air-quality research in the region rather than a one-time artefact. The accompanying code and the reproducible tensor-construction routine make it straightforward to regenerate the dataset and to add districts at finer spatial granularity in future, supporting the long-term evolution of the resource.

5.3 Project Management and Financial Analysis

The project ran in two phases, dataset construction followed by modelling, each occupying roughly half of the project timeline, with the writing carried out alongside the later stages. The detailed week-by-week task allocation is given separately.

Financially, the project was deliberately low-cost, which is itself part of its value. The main resources and their costs are summarised in Table 5.4.

Table 5.1 — Resource and cost summary.

Resource	Source	Cost
Satellite and reanalysis data	Google Earth Engine (open archives)	Free
Ground-truth measurements	OpenAQ and Bangladesh DoE (public)	Free
Independent validation product	ACAG (public)	Free
Software	Python, PyTorch, scikit-learn (open source)	Free
Computation	Single cloud GPU for training	Low (a small amount of GPU time)
Researcher time	Project author(s)	The principal investment

The takeaway is that the dominant cost of this work was human effort, not money or hardware. All the data and tools are open, and the computation is light. This makes the approach affordable to reproduce and extend, which matters for a low-resource setting and reinforces the project's sustainability. Chapter 6 concludes the report.

5.4 Complex Engineering Problem

The project has the attributes of a complex engineering problem. It is presented below in tables against the standard attributes, the knowledge profile and the engineering activities.

Table 5.2 — Complex Engineering Problem attributes.

Attribute	How the project addresses it
EP1 — Depth of knowledge	Requires deep, research-level knowledge spanning atmospheric science, satellite remote sensing, statistics, machine learning, and graph neural networks.

EP2 — Range of conflicting requirements	Balances accuracy against severe data sparsity, spatial coverage against the lack of ground truth, model complexity against overfitting, and impressive numbers against honest evaluation.
EP3 — Depth of analysis	There was no canned solution; the work involved building a dataset, designing a custom three-track evaluation, and using multi-seed significance testing and ablations to reach defensible conclusions.
EP4 — Familiarity of issues	The core issues — zero-shot prediction in unmonitored regions and learning the graph from data — fall well outside standard, textbook problems.
EP5 — Extent of applicable codes	There are not many prescriptive codes for a research problem of this type, but the work follows the WHO guidelines, the EPA AQI standard and FAIR data and reproducibility practices.
EP6 — Stakeholder involvement	Serves diverse stakeholders with differing needs — public-health researchers, environmental regulators, policymakers, and the general public.
EP7 — Interdependence	Highly interdependent: the dataset depends on fusing several data sources, the models depend on the dataset, and the evaluation depends on the provenance tagging.

Table 5.3 — Knowledge profile.

Knowledge area	Applied in this project
WK1 — Natural sciences	Atmospheric science and aerosol behaviour underpinning the AOD-to-PM2.5 relationship.
WK2 — Mathematics and statistics	Significance testing, the attention mechanism, and the calibration model.
WK3 — Engineering fundamentals	Machine-learning and deep-learning fundamentals.
WK4 — Specialist knowledge	Spatiotemporal graph neural networks and satellite remote sensing.
WK5 — Engineering design	Design of the data pipeline, the proposed model, and the evaluation protocol.
WK6 — Engineering practice	Use of modern tools — Google Earth Engine, PyTorch, scikit-learn.
WK7 — Role in society	Direct engagement with public health, the environment, and sustainability.
WK8 — Research literature	An extensive review of, and engagement with, the relevant research literature.

Table 5.4 — Complex engineering activities.

Activity	How the project demonstrates it
A1 — Range of resources	Integrates multiple satellite and reanalysis archives, two ground networks, and an independent validation product, using cloud computing and specialist libraries.
A2 — Level of interaction	Resolves wide-ranging and conflicting technical issues across data, modelling, and evaluation.
A3 — Innovation	Produces a first-of-its-kind dataset, the first regional ST-GNN benchmark, and a learned-adjacency model adapted to a data-sparse setting.
A4 — Consequences for society and environment	Carries significant consequences for public health and environmental policy.
A5 — Familiarity	Goes well beyond routine engineering, requiring research-level judgement throughout.

5.5 Summary

This chapter has discussed the project's commitment to open-data, software, reproducibility and satellite-data standards, and has considered its wider impact: its potential to inform air-quality management and public understanding, its minimal environmental cost, the ethical imperative of honesty regarding the observed-versus-predicted distinction, and a sustainability plan built on open, extensible infrastructure.

Chapter 6

Conclusion

This chapter summarizes the work and its contributions, acknowledges its limitations, and outlines directions for future research.

6.1 Summary

This project was designed to fill two related gaps in the study of air quality in Bangladesh: the lack of a high-resolution, multi-year dataset for PM2.5 and the lack of any spatiotemporal graph neural network modelling for the region. Both were covered in a large body of work. We assembled a daily PM2.5 dataset for all 64 districts from 2019 to 2025 by integrating satellite remote sensing, meteorological reanalysis, and ground measurements from two complementary networks. Random Forest calibration yielded a complete field and a transparent per-record provenance. Validation of the dataset was performed against an independent satellite product and good agreement was found, even in districts that were not monitored. A detailed exploratory analysis in a disciplined separation of measured and predicted values, characterised the country's pollution climatology, correcting the Dhaka-centric narrative, and elucidated the combustion-driven winter-trap mechanism controlling the seasonal cycle.

This dataset has been used by the project to develop the first spatiotemporal graph neural network benchmark for air-quality prediction in Bangladesh, with baseline, sequence, established graph and novel hybrid models evaluated under a three-track evaluation protocol designed for extreme label sparsity, with rigorous multi-seed statistical testing. The central finding is that spatial graph structure improves prediction only in the data-sparse regime: in well-observed districts a tuned sequence model matches every graph model, whereas in unmonitored districts a learned-attention spatial mechanism yields a small but statistically significant improvement. This benefit was shown to require learned rather than fixed adjacency and to be degraded by additional temporal complexity, locating the effective architecture at a specific point of moderate complexity. All models were found to share a common ceiling on zero-shot accuracy, indicating that the binding constraint on prediction is the density of ground-truth monitoring rather than the capacity of the model.

The contributions of the work are therefore threefold: a novel, openly available, validated dataset that did not previously exist; the first ST-GNN benchmark for the region, together with a transferable evaluation methodology for data-sparse spatiotemporal prediction; and a set of rigorously established empirical findings, both positive and negative, about when and why spatial structure helps in this setting.

6.2 Limitation

The work has several limitations, which are stated plainly. The ground-truth data are sparse and heavily biased toward Dhaka, which caps the achievable accuracy of the calibration model and widens the variance of the modeling results; this sparsity is the source of the data ceiling observed in the benchmark. The calibration model is trained in part on its own imputed field, so the models are to some degree shaped by the calibration; this is mitigated, but not eliminated, by masking the training objective to observed cells and by independent validation. The calibration model, being a tree ensemble, regresses toward the mean and cannot reproduce the most extreme pollution events, and it produces temporally smoother series than reality, which is why all analysis and evaluation are confined to observed data. Conversion of government Air Quality Index values to PM2.5 causes compression at extreme concentrations. The dataset aggregates pollution to the district level and does not capture sub-district heterogeneity. Finally, the positive modeling result, while statistically significant, is small in magnitude and is presented as a robust but modest effect rather than a transformative one; and the novel models are adaptations of standard neural primitives rather than fundamentally new architectures.

6.3 Future Work

These limitations and findings suggest several avenues for future work. The most important is the expansion of ground-truth monitoring, particularly outside Dhaka and in the northern and coastal districts identified as least well characterised; since the benchmark identified ground-truth density as the binding constraint, additional measurement would go further to improve prediction than any architectural change. Methodologically, the data-sparse regime beckons to approaches not pursued here: inductive spatio-temporal interpolation (prediction at unmonitored sites, not forecasting); transfer learning from data-rich areas such as the greater Indo-Gangetic Plain, followed by fine tuning on Bangladeshi data; semi-supervised goals that make better use of the unlabelled field than the auxiliary-loss method used here. Uncertainty measurement is adjusted against the independent reference product, which allows predictions in unmonitored areas to have reliable confidence intervals. Expanding the dataset to a more detailed spatial level and updating it as new data is collected would further enhance its usefulness as a lasting resource for air quality research in the region.

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